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Introduction to Embedded Systems

LEARNING OBJECTIVES

- LO 1 Know what an embedded system is
- LO 2 Differentiate between embedded and general computing systems
- LO 3 Underline the history of embedded systems
- LO 4 Classify embedded systems based on performance, complexity and the era in which they evolved
- LO 5 Explain the domains and areas of applications of embedded systems
- LO 6 Identify the different purposes of embedded systems
- LO 7 Analyse a real life example on the bonding of embedded technology with human life

Our day-to-day life is becoming more and more dependent on “embedded systems” and digital technologies. Embedded technologies are bonding into our daily activities even without our knowledge. Do you know the fact that the refrigerator, washing machine, microwave oven, air conditioner, television, DVD players, and music systems that we use in our home are built around an embedded system? You may be traveling by a ‘Honda’ or a ‘Toyota’ or a ‘Ford’ vehicle, but have you ever thought of the genius players working behind the special features and security systems offered by the vehicle to you? It is nothing but an intelligent embedded system. In your vehicle itself the presence of specialised embedded systems vary from intelligent head lamp controllers, engine controllers and ignition control systems to complex air bag control systems to protect you from a severe accident. People experience the power of embedded systems and enjoy the features and comfort provided by them. Most of us are totally unaware or ignorant of the intelligent embedded systems giving us so much comfort and security. Embedded systems are like reliable servants—they don’t like to reveal their identity and neither they complain about their workloads to their owners or bosses. They are always working behind the scenes and are dedicated to their assigned task till their last breath. This book gives you an overview of embedded systems, the various steps involved in their design and development and the major domains where they are deployed.

1.1 WHAT IS AN EMBEDDED SYSTEM?

LO 1 Know what an embedded system is

An embedded system is an electronic/electro-mechanical system designed to perform a specific function and is a combination of both hardware and firmware (software).

Every embedded system is unique, and the hardware as well as the firmware is highly specialised to the application domain. Embedded systems are becoming an inevitable part of any product or equipment in all fields including household appliances, telecommunications, medical equipment, industrial control, consumer products, etc.

1.2 EMBEDDED SYSTEMS vs. GENERAL COMPUTING SYSTEMS

LO 2 Differentiate between embedded and general computing systems

The computing revolution began with the general purpose computing requirements. Later it was realised that the general computing requirements are not sufficient for the embedded computing requirements. The embedded computing requirements demand ‘something special’ in terms of response to stimuli, meeting the computational deadlines, power efficiency, limited memory availability, etc.

Let’s take the case of your personal computer, which may be either a desktop PC or a laptop PC or a tablet PC. It is built around a general purpose processor like an Intel® Celeron/Core M or a Duo/Quad* core or an AMD A-Series processor and is designed to support a set of multiple peripherals like multiple USB 3.0 ports, Wi-Fi, ethernet, video port, IEEE1394, SD/CF/MMC external interfaces, Bluetooth, etc and with additional interfaces like a DVD read/writer, on-board Hard Disk Drive (HDD), gigabytes of RAM, etc. You can load any supported operating system (like Windows® 8.X/10, or Red Hat Linux/Ubuntu Linux, UNIX etc) into the hard disk of your PC. You can write or purchase a multitude of applications for your PC and can use your PC for running a large number of applications (like printing your dear’s photo using a printer device connected to the PC and printer software, creating a document using Microsoft® Office Word tool, etc.) Now, let us think about the DVD player you use for playing DVD movies. Is it possible for you to change the operating system of your DVD? Is it possible for you to write an application and download it to your DVD player for executing? Is it possible for you to add a printer software to your DVD player and connect a printer to your DVD player to take a printout? Is it possible for you to change the functioning of your DVD player to a television by changing the embedded software? The answers to all these questions are ‘NO’. Can you see any general purpose interface like Bluetooth or Wi-Fi on your DVD player? Of course ‘NO’. The only interface you can find out on the DVD player is the interface for connecting the DVD player with the display screen and one for controlling the DVD player through a remote (May be an IR or any other specific wireless interface). Indeed your DVD player is an embedded system designed specifically for decoding digital video and generating a video signal as output to your TV or any other display screen which supports the display interface supported by the DVD Player. Let us summarise our findings from the comparison of embedded system and general purpose computing system with the help of a table:

General Purpose Computing System	Embedded System
A system which is a combination of a generic hardware and a General Purpose Operating System for executing a variety of applications	A system which is a combination of special purpose hardware and embedded OS for executing a specific set of applications

(Contd.)

*The illustration given here is based on the processor details available till Dec 2015. Since processor technology is undergoing rapid changes, the processor names mentioned here may not be relevant in future.

General Purpose Computing System	Embedded System
Contains a General Purpose Operating System (GPOS)	May or may not contain an operating system for functioning
Applications are alterable (programmable) by the user (It is possible for the end user to re-install the operating system, and also add or remove user applications)	The firmware of the embedded system is pre-programmed and it is non-alterable by the end-user (There may be exceptions for systems supporting OS kernel image flashing through special hardware settings)
Performance is the key deciding factor in the selection of the system. Always, 'Faster is Better'	Application-specific requirements (like performance, power requirements, memory usage, etc.) are the key deciding factors
Less/not at all tailored towards reduced operating power requirements, options for different levels of power management.	Highly tailored to take advantage of the power saving modes supported by the hardware and the operating system
Response requirements are not time-critical	For certain category of embedded systems like mission critical systems, the response time requirement is highly critical
Need not be deterministic in execution behaviour	Execution behaviour is deterministic for certain types of embedded systems like 'Hard Real Time' systems

However, the demarcation between desktop systems and embedded systems in certain areas of embedded applications are shrinking in certain contexts. Smart phones are typical examples of this. Nowadays smart phones are available with RAM 2 to 3 GB and users can extend most of their desktop applications to the smart phones and it invalidates the clause "Embedded systems are designed for a specific application" from the characteristics of the embedded system for the mobile embedded device category. However, smart phones come with a built-in operating system and it is not modifiable by the end user. It makes the clause: "The firmware of the embedded system is unalterable by the end user", still a valid clause in the mobile embedded device category.

1.3 HISTORY OF EMBEDDED SYSTEMS

Embedded systems were in existence even before the IT revolution. In the olden days, embedded systems were built around the old vacuum tube and transistor technologies and the embedded algorithm was developed in low level languages. Advances in semiconductor and nano-technology and IT revolution gave way to the development of miniature embedded systems. The first recognised modern embedded system is the Apollo Guidance Computer (AGC) developed by the MIT Instrumentation Laboratory for the lunar expedition. They ran the inertial guidance systems of both the Command Module (CM) and the Lunar Excursion Module (LEM). The Command Module was designed to encircle the moon while the Lunar Module and its crew were designed to go down to the moon surface and land there safely. The Lunar Module featured in total 18 engines. There were 16 reaction control thrusters, a descent engine and an ascent engine. The descent engine was 'designed to' provide thrust to the lunar module out of the lunar orbit and land it safely on the moon. MIT's original design was based on 4K words of fixed memory (Read Only Memory) and 256 words of erasable memory (Random Access Memory). By June 1963, the figures reached 10K of fixed and 1K of erasable memory. The final configuration was 36K words of fixed memory and 2K words of erasable memory. The clock frequency of the first microchip proto model used in AGC

**LO 3 Underline
the history of
embedded systems**

was 1.024 MHz and it was derived from a 2.048 MHz crystal clock. The computing unit of AGC consisted of approximately 11 instructions and 16 bit word logic. Around 5000 ICs (3-input NOR gates, RTL logic) supplied by Fairchild Semiconductor were used in this design. The user interface unit of AGC is known as DSKY (display/keyboard). DSKY looked like a calculator type keypad with an array of numerals. It was used for inputting the commands to the module numerically.

The first mass-produced embedded system was the guidance computer for the Minuteman-I missile in 1961. It was the ‘Autonetics **D-17**’ guidance computer, built using discrete transistor logic and a hard-disk for main memory. The first integrated circuit was produced in September 1958 but computers using them didn’t begin to appear until 1963. Some of their early uses were in embedded systems, notably used by NASA for the Apollo Guidance Computer and by the US military in the Minuteman-II intercontinental ballistic missile.

1.4 CLASSIFICATION OF EMBEDDED SYSTEMS

LO 4 Classify embedded systems based on performance, complexity and the era in which they evolved

It is possible to have a multitude of classifications for embedded systems, based on different criteria. Some of the criteria used in the classification of embedded systems are as follows:

- (1) Based on generation
- (2) Complexity and performance requirements
- (3) Based on deterministic behaviour
- (4) Based on triggering.

The classification based on deterministic system behaviour is applicable for ‘Real Time’ systems. The application/task execution behaviour for an embedded system can be either deterministic or non- deterministic. Based on the execution behaviour, Real Time embedded systems are classified into *Hard* and *Soft*. We will discuss about hard and soft real time systems in a later chapter. Embedded Systems which are ‘Reactive’ in nature (Like process control systems in industrial control applications) can be classified based on the trigger. Reactive systems can be either *event triggered* or *time triggered*.

1.4.1 Classification Based on Generation

This classification is based on the order in which the embedded processing systems evolved from the first version to where they are today. As per this criterion, embedded systems can be classified into the following:

1.4.1.1 First Generation The early embedded systems were built around 8bit microprocessors like 8085 and Z80, and 4bit microcontrollers. Simple in hardware circuits with firmware developed in Assembly code. Digital telephone keypads, stepper motor control units etc. are examples of this.

1.4.1.2 Second Generation These are embedded systems built around 16bit microprocessors and 8 or 16 bit microcontrollers, following the first generation embedded systems. The instruction set for the second generation processors/controllers were much more complex and powerful than the first generation processors/controllers. Some of the second generation embedded systems contained embedded operating systems for their operation. Data Acquisition Systems, SCADA systems, etc. are examples of second generation embedded systems.

1.4.1.3 Third Generation With advances in processor technology, embedded system developers started making use of powerful 32bit processors and 16bit microcontrollers for their design. A new concept of

application and domain specific processors/controllers like Digital Signal Processors (DSP) and Application Specific Integrated Circuits (ASICs) came into the picture. The instruction set of processors became more complex and powerful and the concept of instruction pipelining also evolved. The processor market was flooded with different types of processors from different vendors. Processors like Intel Pentium, Motorola 68K, etc. gained attention in high performance embedded requirements. Dedicated embedded real time and general purpose operating systems entered into the embedded market. Embedded systems spread its ground to areas like robotics, media, industrial process control, networking, etc.

1.4.1.4 Fourth Generation The advent of System on Chips (SoC), reconfigurable processors and multicore processors are bringing high performance, tight integration and miniaturisation into the embedded device market. The SoC technique implements a total system on a chip by integrating different functionalities with a processor core on an integrated circuit. We will discuss about SoCs in a later chapter. The fourth generation embedded systems are making use of high performance real time embedded operating systems for their functioning. Smart phone devices, mobile internet devices (MIDs), etc. are examples of fourth generation embedded systems.

1.4.1.5 What Next? The processor and embedded market is highly dynamic and demanding. So ‘what will be the next smart move in the next embedded generation?’ Let’s wait and see.

1.4.2 Classification Based on Complexity and Performance

This classification is based on the complexity and system performance requirements. According to this classification, embedded systems can be grouped into the following:

1.4.2.1 Small-Scale Embedded Systems Embedded systems which are simple in application needs and where the performance requirements are not time critical fall under this category. An electronic toy is a typical example of a small-scale embedded system. Small-scale embedded systems are usually built around low performance and low cost 8 or 16 bit microprocessors/microcontrollers. A small-scale embedded system may or may not contain an operating system for its functioning.

1.4.2.2 Medium-Scale Embedded Systems Embedded systems which are slightly complex in hardware and firmware (software) requirements fall under this category. Medium-scale embedded systems are usually built around medium performance, low cost 16 or 32 bit microprocessors/microcontrollers or digital signal processors. They usually contain an embedded operating system (either general purpose or real time operating system) for functioning.

1.4.2.3 Large-Scale Embedded Systems/Complex Systems Embedded systems which involve highly complex hardware and firmware requirements fall under this category. They are employed in mission critical applications demanding high performance. Such systems are commonly built around high performance 32 or 64 bit RISC processors/controllers or Reconfigurable System on Chip (RSoC) or multi-core processors and programmable logic devices. They may contain multiple processors/controllers and co-units/hardware accelerators for offloading the processing requirements from the main processor of the system. Decoding/encoding of media, cryptographic function implementation, etc. are examples for processing requirements which can be implemented using a co-processor/hardware accelerator. Complex embedded systems usually contain a high performance Real Time Operating System (RTOS) for task scheduling, prioritisation, and management.

1.5 MAJOR APPLICATION AREAS OF EMBEDDED SYSTEMS

LO 5 Explain the domains and areas of applications of embedded systems

We are living in a world where embedded systems play a vital role in our day-to-day life, starting from home to the computer industry, where most of the people find their job for a livelihood. Embedded technology has acquired a new dimension from its first generation model, the Apollo guidance computer, to the latest radio navigation system combined with in-car entertainment technology and the wearable computing devices (Apple watch, Microsoft Band, Fitbit fitness trackers etc.). The application areas and the products in the embedded domain are countless. A few of the important domains and products are listed below:

- (1) *Consumer electronics*: Camcorders, cameras, etc.
- (2) *Household appliances*: Television, DVD players, washing machine, fridge, microwave oven, etc.
- (3) *Home automation and security systems*: Air conditioners, sprinklers, intruder detection alarms, closed circuit television cameras, fire alarms, etc.
- (4) *Automotive industry*: Anti-lock breaking systems (ABS), engine control, ignition systems, automatic navigation systems, etc.
- (5) *Telecom*: Cellular telephones, telephone switches, handset multimedia applications, etc.
- (6) *Computer peripherals*: Printers, scanners, fax machines, etc.
- (7) *Computer networking systems*: Network routers, switches, hubs, firewalls, etc.
- (8) *Healthcare*: Different kinds of scanners, EEG, ECG machines etc.
- (9) *Measurement & Instrumentation*: Digital multimeters, digital CROs, logic analysers PLC systems, etc.
- (10) *Banking & Retail*: Automatic teller machines (ATM) and currency counters, point of sales (POS)
- (11) *Card Readers*: Barcode, smart card readers, hand held devices, etc.
- (12) *Wearable Devices*: Health and Fitness Trackers, Smartphone Screen extension for notifications, etc.
- (13) Cloud Computing and Internet of Things (IOT)

1.6 PURPOSE OF EMBEDDED SYSTEMS

LO 6 Identify the different purposes of embedded systems

As mentioned in the previous section, embedded systems are used in various domains like consumer electronics, home automation, telecommunications, automotive industry, healthcare, control & instrumentation, retail and banking applications, etc. Within the domain itself, according to the application usage context, they may have different functionalities. Each embedded system is designed to serve the purpose of any one or a combination of the following tasks:

- (1) Data collection/Storage/Representation
- (2) Data communication
- (3) Data (signal) processing
- (4) Monitoring
- (5) Control
- (6) Application specific user interface

1.6.1 Data Collection/Storage/Representation

Embedded systems designed for the purpose of data collection performs acquisition of data from the external world. Data collection is usually done for storage, analysis, manipulation, and transmission. The term “data” refers to all kinds of information, viz. text, voice, image, video, electrical signals and any other measurable quantities. Data can be either analog (continuous) or digital (discrete). Embedded systems with analog data capturing techniques collect data directly in the form of analog signals whereas embedded systems with digital data collection mechanism converts the analog signal to corresponding digital signal using analog to digital (A/D) converters and then collects the binary equivalent of the analog data. If the data is digital, it can be directly captured without any additional interface by digital embedded systems.

The collected data may be stored directly in the system or may be transmitted to some other systems or it may be processed by the system or it may be deleted instantly after giving a meaningful representation. These actions are purely dependent on the purpose for which the embedded system is designed. Embedded systems designed for pure measurement applications without storage, used in control and instrumentation domain, collect data and give a meaningful representation of the collected data by means of graphical representation or quantity value and delete the collected data when new data arrives at the data collection terminal. Analog and digital CROs without storage memory are typical examples of this. Any measuring equipment used in the medical domain for monitoring without storage functionality also comes under this category.

Some embedded systems store the collected data for processing and analysis. Such systems incorporate a built-in/plugin storage memory for storing the captured data. Some of them give the user a meaningful representation of the collected data by visual (graphical/quantitative) or audible means using display units [Liquid Crystal Display (LCD), Light Emitting Diode (LED), etc.] buzzers, alarms, etc. Examples are: measuring instruments with storage memory and monitoring instruments with storage memory used in medical applications. Certain embedded systems store the data and will not give a representation of the same to the user, whereas the data is used for internal processing.

A digital camera is a typical example of an embedded system with data collection/storage/representation of data. Images are captured and the captured image may be stored within the memory of the camera. The captured image can also be presented to the user through a graphic LCD unit.



Fig. 1.1 A digital camera for image capturing/ storage/display
(Photo courtesy of Casio-Model EXILIM ex-Z850
(www.casio.com))

1.6.2 Data Communication

Embedded data communication systems are deployed in applications ranging from complex satellite communication systems to simple home networking systems. As mentioned earlier in this chapter, the data collected by an embedded terminal may require transferring of the same to some other system located remotely. The transmission is achieved either by a wire-line medium or by a wireless medium. Wire-line medium was the most common choice in all olden days embedded systems. As technology is changing, wireless medium is becoming the de-facto standard for data communication in embedded systems. A wireless medium offers cheaper connectivity solutions and make the communication link free from the hassle of wire bundles. Data can either be transmitted by analog means or by digital means. Modern industry trends are settling towards digital communication.

The data collecting embedded terminal itself can incorporate data communication units like wireless modules (Bluetooth, ZigBee, Wi-Fi, EDGE, GPRS, etc.) or wire-line modules (RS-232C, USB, TCP/IP, PS2, etc.). Certain embedded systems



Fig. 1.2 A wireless network router for data communication (Photo courtesy of Linksys
(www.linksys.com). A division of CISCO system)

act as a dedicated transmission unit between the sending and receiving terminals, offering sophisticated functionalities like data packetizing, encrypting and decrypting. Network hubs, routers, switches, etc. are typical examples of dedicated data transmission embedded systems. They act as mediators in data communication and provide various features like data security, monitoring etc.

1.6.3 Data (Signal) Processing

As mentioned earlier, the data (voice, image, video, electrical signals, and other measurable quantities) collected by embedded systems may be used for various kinds of data processing. Embedded systems with signal processing functionalities are employed in applications demanding signal processing like speech coding, synthesis, audio video codec, transmission applications, etc.

A digital hearing aid is a typical example of an embedded system employing data processing. Digital hearing aid improves the hearing capacity of hearing impaired persons.

1.6.4 Monitoring

Embedded systems falling under this category are specifically designed for monitoring purpose. Almost all embedded products coming under the medical domain are with monitoring functions only. They are used for determining the state of some variables using input sensors. They cannot impose control over variables. A very good example is the electro cardiogram (ECG) machine for monitoring the heartbeat of a patient. The machine is intended to do the monitoring of the heartbeat. It cannot impose control over the heartbeat. The sensors used in ECG are the different electrodes connected to the patient's body.

Some other examples of embedded systems with monitoring function are measuring instruments like digital CRO, digital multimeters, logic analysers, etc. used in Control & Instrumentation applications. They are used for knowing (monitoring) the status of some variables like current, voltage, etc. They cannot control the variables in turn.

1.6.5 Control

Embedded systems with control functionalities impose control over some variables according to the changes in input variables. A system with control functionality contains both sensors and actuators. Sensors are connected to the input port for capturing the changes in environmental variable or measuring variable. The

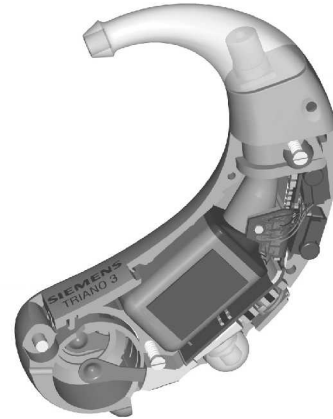


Fig. 1.3 A digital hearing aid employing signal processing technique (Siemens TRIANO 3 Digital hearing aid; Siemens Audiology Copyright © 2005)

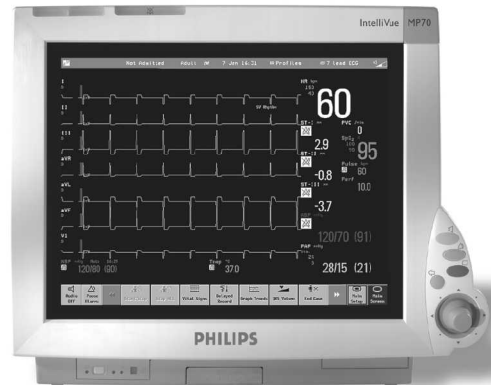


Fig. 1.4 A patient monitoring system for monitoring heartbeat (Photo courtesy of Philips Medical Systems (www.medical.philips.com/))

actuators connected to the output port are controlled according to the changes in input variable to put an impact on the controlling variable to bring the controlled variable to the specified range.

Air conditioner system used in our home to control the room temperature to a specified limit is a typical example for embedded system for control purpose. An air conditioner contains a room temperature-sensing element (sensor) which may be a thermistor and a handheld unit for setting up (feeding) the desired temperature. The handheld unit may be connected to the central embedded unit residing inside the air conditioner through a wireless link or through a wired link. The air compressor unit acts as the actuator. The compressor is controlled according to the current room temperature and the desired temperature set by the end user.

Here the input variable is the current room temperature and the controlled variable is also the room temperature. The controlling variable is cool air flow by the compressor unit. If the controlled variable and input variable are not at the same value, the controlling variable tries to equalise them through taking actions on the cool air flow.

1.6.6 Application Specific User Interface

These are embedded systems with application-specific user interfaces like buttons, switches, keypad, lights, bells, display units, etc. Mobile phone is an example for this. In mobile phone the user interface is provided through the keypad, graphic LCD module, system speaker, vibration alert, etc.



ESG21HRIA

Fig. 1.5 An Air conditioner for controlling room temperature. Embedded System with Control functionality

(Photo courtesy of Electrolux Corporation)



Fig. 1.6 An embedded system with an application-specific user interface

(Photo courtesy of BlackBerry Smartphone Handsets (amazon.com))

1.7 WEARABLE DEVICES—THE INNOVATIVE BONDING OF LIFESTYLE WITH EMBEDDED TECHNOLOGIES

Wearable Devices/Wearables is a hot topic of discussion at the time of the first revision of this book (Year 2015). *Wearable Devices/Wearables* refer to embedded technologies/systems which are incorporated into accessories (watch, bracelet, ring, necklace, eye glass, headband, beanies, etc.) and apparels. Wearable technology envisions the bonding of embedded technologies into our day-to-day life and is setting a new dimension to fashion and lifestyle. It empowers you with the power of embedded computing and communication, keeping you always connected and alerted with things which you are most interested (e.g. Social Feeds - Twitter, Facebook, etc., Instant Messages from your dear and near, tracking your fitness activities, etc.), without compromising on the fashion aspects of your lifestyle.

LO 7 Analyse a real life example on the bonding of embedded technology with human life

The calculator watch from Casio, introduced in the 1980s, was a classic example for the concept of wearable device. In terms of the computing power, today's wearable devices are capable of performing many of the same computing tasks as smartphones and tablets; and in some cases they can even outperform these gadgets. While the term *Wearables* is generally used to specify a set of accessories and apparels which can be put on and taken off easily, certain type of devices which are implanted into the body like smart tattoos, RFID microchips etc. also falls under the broad definition of *Wearables*. The primary goal of wearable technologies is to smoothly integrate functional, portable embedded computing devices into individuals' day-to-day life to provide constant, convenient, seamless integration (Information at a quick glance—View your social feeds from Facebook, Twitter, etc. on the screen of a Smartwatch which is connected to your Smartphone and act as a second screen for your phone) and hands-free access to other connected devices (e.g. Interact with a gaming console or Music player App on a smartphone through simulated touch gestures from a wearable device like smart ring). The majority of the wearable devices out there in the market today/proposed for future fall under the Health and Fitness monitoring/Entertainment/Smartphone Accessory or feature extension category.

What is inside a wearable device is solely dependent on the targeted end functionality. In general, wearable device includes an embedded controller which acts as the master brain of the unit, a wireless Communication subsystem [(Based on Wireless technologies like Bluetooth/Wi-Fi/NFC (Near Field Communication)/Cellular, etc.)] for establishing data/voice communication with other electronic systems like Smartphones, Laptops, Tablets, Gaming consoles, etc. Let us consider a concept Smartwatch which acts a normal watch and at the same time has a bunch of sensors embedded into it to keep track of the various fitness activities like number of steps taken and distance walked (Pedometer functionality), number of floors climbed etc. and to track and record vital signs like heart-rate. An embedded Processor/Microcontroller (Like ARM) will act as the master controller unit, which interfaces with the on-board subsystems like sensor modules which keep track of the various fitness activities and record the vital signals, wireless communication link like Bluetooth for establishing communication channel with an interfacing device like Smartphone/Tablet. The pedometer functionality is implemented through MEMS 3-D accelerometer sensor (e.g. LIS2DH from ST Microelectronics). An Altimeter/Barometric sensor can be used for tracking the number of floors climbed. The heart rate monitoring functionality is implemented by the combination of an Optic/Photo sensor or Pressure Sensor or Electrodes with an Analog Front End (AFE) chip (e.g. Analog Devices AD8232 AFE chip) for signal conditioning. Depending on the volume of readings to be stored in the device, the device may contain a few kilobytes to a few Megabytes of Flash (Storage) memory. An ultra-low power Bluetooth transceiver is used for establishing a wireless communication interface with the device and it can talk to other peripherals over standard Bluetooth profiles like (SPP - Serial Port Profile, MAP - Message Access Profile, GATT - Generic Attribute Profile, etc.). Since the device is intended to work as a standalone, portable device it will contain a battery (Mostly a rechargeable Li-Io battery) to power the unit and a battery charging and monitoring subsystem to re-charge the battery and to keep track of the battery status.

Fitbit Charge, Surge (www.fitbit.com), are examples of wearable devices for fitness and activity tracking. Sony SmartWatch 3 (<http://www.sonymobile.com/us/products/smartwear/smartwatch-3-swr50/>), Samsung Galaxy Gear Series Smart watches are examples of wearable devices which provide the functionality of a watch and at the same time acts an accessory/extension to your smartphone - Featuring Quick Glance notifications, such as texts, emails social feeds, phone calls, hands-free calling all on your wrist. APPLE WATCH announced in Sep 2014 is an example of a Wearable device which acts as an accessory to the iPhone for providing notifications and content from apps and also integrates the functionality of a watch, health and fitness tracking, mobile payment and many other niche features. Microsoft Band is a wearable device which provides integration with multi-mobile-platforms (Windows/Android/iOS) for notifications (Call, Text, Social Feeds etc.) and also acts a fitness tracker and health monitoring device. It integrates multiple sensors (Optical Heart Rate sensor, 3-axis accelerometer/gyro, GPS, Ambient light sensor, Skin temperature sensor, UV sensor, Galvanic skin response sensor, Barometer etc.) and tracks steps, heart rate,

calories burned, sleep quality, floors climbed, detects the presence of harmful UV light and maps the route during a run.

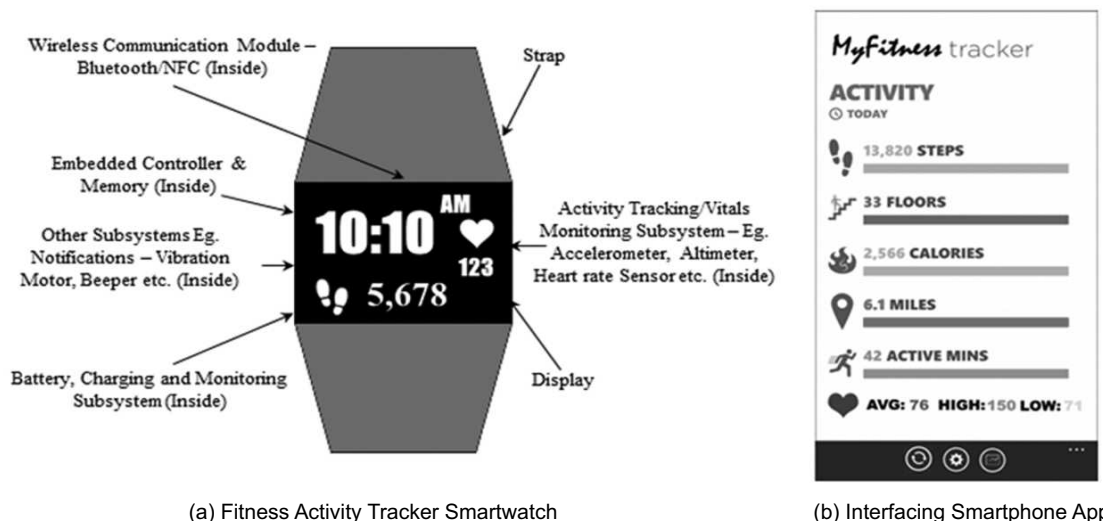


Fig. 1.7 Fitness Watch—A concept Model of a Wearable Device for Fitness Activity Tracking

The wearable technology is constantly evolving and gaining more popularity. It has the potential to significantly influence the fields of Interactive Entertainment (Gaming, Music, Video etc.) Health and Fitness, Education, Enterprise, Transportation, Retail etc. According to a leading US Market analysis and Research firm, the total shipment volume of wearable devices is expected to surpass 150 millions by year 2019 with a Compound Annual Growth Rate (CGAR) of over 40%.

Summary

- ✓ An embedded system is an Electronic/Electro-mechanical system designed to perform a specific function and is a combination of both hardware and firmware (Software). **LO1**
- ✓ A general purpose computing system is a combination of generic hardware and general purpose operating system for executing a variety of applications, whereas an embedded system is a combination of special purpose hardware and embedded OS/firmware for executing a specific set of applications. **LO2**
- ✓ Apollo Guidance Computer (AGC) is the first recognised modern embedded system and 'Autonetics D-17', the guidance computer for the Minuteman-I missile, was the first mass-produced embedded system. **LO3**
- ✓ Based on the complexity and performance requirements, embedded systems are classified into small-scale, medium-scale and large-scale/complex. **LO4**
- ✓ The presence of embedded systems vary from simple electronic toys to complex flight and missile control systems. **LO5**
- ✓ Embedded systems are designed to serve the purpose of any one or a combination of data collection/storage/representation, data communication, data (signal) processing, monitoring, control or application specific user interface. **LO6**
- ✓ Wearable devices refer to embedded technologies/systems which are incorporated into accessories and apparels. It envisions the bonding of embedded technology in our day-to-day lives. **LO7**

Keywords

Embedded system: An electronic/electro-mechanical system which is designed to perform a specific function and is a combination of both hardware and firmware	[LO 1]
Operating system: A piece of software designed to manage and allocate system resources and execute other pieces of software	[LO 2]
RAM: Random Access memory. Volatile memory	[LO 3]
Microprocessor: A silicon chip representing a Central Processing Unit (CPU)	[LO 4]
Microcontroller: A highly integrated chip that contains a CPU, scratchpad RAM, special and general purpose register arrays and integrated peripherals	[LO 4]
SCADA: Supervisory Control and Data Acquisition System. A data acquisition system used in industrial control applications	[LO 4]
DSP: Digital Signal Processor is a powerful special purpose 8/16/32 bit microprocessor designed specifically to meet the computational demands and power constraints	[LO 5]
ASIC: Application Specific Integrated Circuit is a microchip designed to perform a specific or unique application	[LO 6]
Sensor: A transducer device that converts energy from one form to another for any measurement or control purpose	[LO 6]
Actuator: A form of transducer device (mechanical or electrical) which converts signals to corresponding physical action (motion)	[LO 6]
LED: Light Emitting Diode. An output device producing visual indication in the form of light in accordance with current flow	[LO 6]
Buzzer: A piezo-electric device for generating audio indication. It contains a piezo-electric diaphragm which produces audible sound in response to the voltage applied to it	[LO 6]
Electro Cardiogram (ECG): An embedded device for heartbeat monitoring	[LO 6]
ADC: Analog to Digital Converter. An integrated circuit which converts analog signals to digital form	[LO 6]
Bluetooth: A low cost, low power, short range wireless technology for data and voice communication	[LO 6]
Wi-Fi: Wireless Fidelity is the popular wireless communication technique for networked communication of devices	[LO 6]

Objective Questions

- Embedded systems are
 - General purpose
 - Special purpose
- Embedded system is
 - An electronic system
 - A pure mechanical system
 - An electro-mechanical system
 - (a) or (c)
- Which of the following is not true about embedded systems?
 - Built around specialised hardware
 - Always contain an operating system
 - Execution behaviour may be deterministic
 - All of these
 - None of these

4. Which of the following is not an example of a 'Small-scale Embedded System'?
 - (a) Electronic Barbie doll
 - (b) Simple calculator
 - (c) Cell phone
 - (d) Electronic toy car
5. The first recognised modern embedded system is
 - (a) Apple Computer
 - (b) Apollo Guidance Computer (AGC)
 - (c) Calculator
 - (d) Radio Navigation System
6. The first mass produced embedded system is
 - (a) Minuteman-I
 - (b) Minuteman-II
 - (c) Autonetics D-17
 - (d) Apollo Guidance Computer (AGC)
7. Which of the following is (are) an intended purpose(s) of embedded systems?
 - (a) Data collection
 - (b) Data processing
 - (c) Data communication
 - (d) All of these
 - (e) None of these
8. Which of the following is an (are) example(s) of embedded system for data communication?
 - (a) USB Mass storage device
 - (b) Network router
 - (c) Digital camera
 - (d) Music player
 - (e) All of these
 - (f) None of these
9. A digital multi meter is an example of an embedded system for
 - (a) Data communication
 - (b) Monitoring
 - (c) Control
 - (d) All of these
 - (e) None of these
10. Which of the following is an (are) example(s) of an embedded system for signal processing?
 - (a) Apple iPod (media player device)
 - (b) SanDisk USB mass storage device
 - (c) Both (a) and (b)
 - (d) None of these

Review Questions

1. What is an embedded system? Explain the different applications of embedded systems.
[LO 1, LO 2, LO 3]
2. Explain the different classifications of embedded systems. Give an example for each.
[LO 4, LO 5, LO 6, LO7]
3. Explain the various purposes of embedded systems in detail with illustrative examples.
[LO 5, LO 6, LO 7]

2

The Typical Embedded System

LEARNING OBJECTIVES

- LO 1 Identify the building blocks of a typical Embedded System
 - ◆ Learn about GPPs, ASIPs, Microprocessors, Microcontrollers, Digital Signal Processors, RISC & CISC processors, Harvard and Von-Neumann Processor Architecture, Big-endian v/s Little endian processors, Load Store operation and Instruction pipelining
 - ◆ Understand different PLDs like CPLDs, FPGAs, etc.
- LO 2 Explain different memory technologies and memory types used in embedded system development
 - ◆ Learn about MROM, PROM, OTP, EPROM, EEPROM and FLASH memory
 - ◆ Describe SAM, SRAM, DRAM, and NVRAM
 - ◆ List the factors to be considered in the memory selection
- LO 3 Analyse the role of sensors, actuators, and their interfacing with the I/O subsystems of an embedded system
 - ◆ Learn about the interfacing of LEDs, 7-segment LED Displays, Piezo Buzzer, Stepper Motor, Relays, Optocouplers, Matrix keyboard, Push button switches, Programmable Peripheral Interface Device (e.g. 8255 PPI), etc. with the I/O subsystem of the embedded system
- LO 4 Examine the different communication interfaces of an embedded system
 - ◆ Understand the various chip level communication interfaces like I2C, SPI, UART, 1-wire, parallel bus, etc.
 - ◆ Know the different wired and wireless external communication interfaces like RS-232C, RS-485, Parallel Port, USB, IEEE 1394, Infrared (IrDA), Bluetooth, Wi-Fi, ZigBee, GPRS, etc.
- LO 5 Describe what embedded firmware is and its role in embedded systems
- LO 6 Know the different system components like Reset Circuit, Brown-out protection circuit, Oscillator Unit, Real-Time Clock (RTC) and Watchdog Timer unit
- LO 7 Discuss the role of PCB in embedded systems

A typical embedded system (Fig. 2.1) contains a single chip controller, which acts as the master brain of the system. The controller can be a Microprocessor (e.g. *Intel 8085*) or a microcontroller (e.g. *Atmel AT89C51*) or a Field Programmable Gate Array (FPGA) device (e.g. *Xilinx Spartan*) or a Digital Signal Processor (DSP) (e.g. *Blackfin® Processors* from Analog Devices) or an Application Specific Integrated Circuit (ASIC)/ Application Specific Standard Product (ASSP) (e.g. *ADE7760 Single Phase Energy Metreing IC* from Analog Devices for energy metering applications).

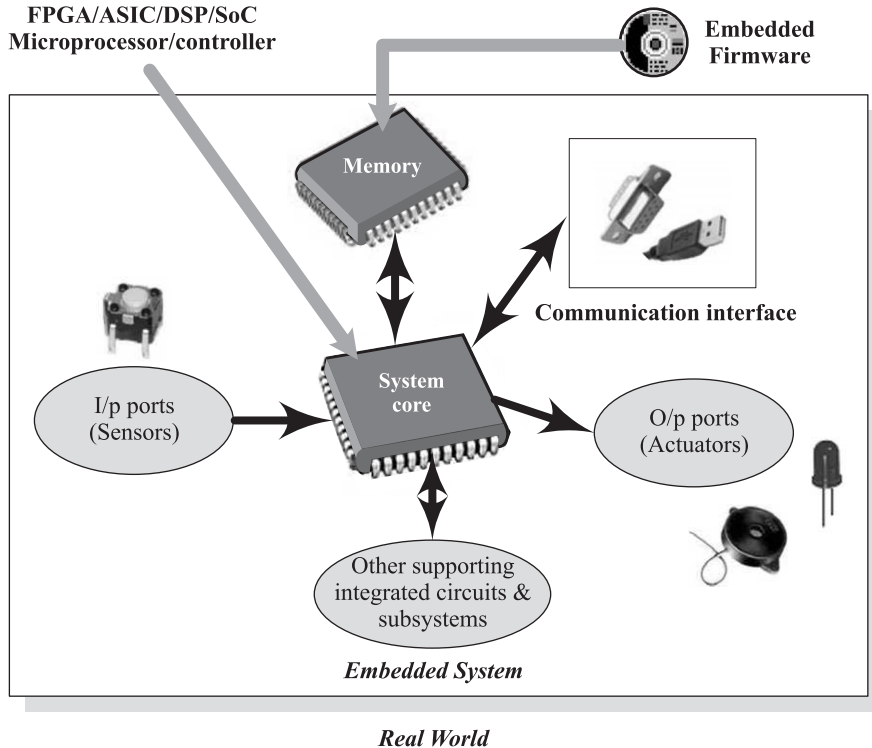


Fig. 2.1 Elements of an embedded system

Embedded hardware/software systems are basically designed to regulate a physical variable or to manipulate the state of some devices by sending some control signals to the Actuators or devices connected to the O/p ports of the system, in response to the input signals provided by the end users or Sensors which are connected to the input ports. Hence an embedded system can be viewed as a reactive system. The control is achieved by processing the information coming from the sensors and user interfaces, and controlling some actuators that regulate the physical variable.

Key boards, push button switches, etc. are examples for common user interface input devices whereas LEDs, liquid crystal displays, piezoelectric buzzers, etc. are examples for common user interface output devices for a typical embedded system. It should be noted that it is not necessary that all embedded systems should incorporate these I/O user interfaces. It solely depends on the type of the application for which the embedded system is designed. For example, if the embedded system is designed for any handheld application, such as a mobile handset application, then the system should contain user interfaces like a keyboard for performing input operations and display unit for providing users the status of various activities in progress.

Some embedded systems do not require any manual intervention for their operation. They automatically sense the variations in the input parameters in accordance with the changes in the real world, to which they are interacting through the sensors which are connected to the input port of the system. The sensor information is passed to the processor after signal conditioning and digitisation. Upon receiving the sensor data the processor or brain of the embedded system performs some predefined operations with the help of the firmware embedded in the system and sends some actuating signals to the actuator connected to the output port of the embedded system, which in turn acts on the controlling variable to bring the controlled variable to the desired level to make the embedded system work in the desired manner.

The Memory of the system is responsible for holding the control algorithm and other important configuration details. For most of embedded systems, the memory for storing the algorithm or configuration data is of fixed type, which is a kind of Read Only Memory (ROM) and it is not available for the end user for modifications, which means the memory is protected from unwanted user interaction by implementing some kind of memory protection mechanism. The most common types of memories used in embedded systems for control algorithm storage are OTP, PROM, UVEPROM, EEPROM and FLASH. Depending on the control application, the memory size may vary from a few bytes to megabytes. We will discuss them in detail in the coming sections. Sometimes the system requires temporary memory for performing arithmetic operations or control algorithm execution and this type of memory is known as “working memory”. Random Access Memory (RAM) is used in most of the systems as the working memory. Various types of RAM like SRAM, DRAM, and NVRAM are used for this purpose. The size of the RAM also varies from a few bytes to kilobytes or megabytes depending on the application. The details given under the section “Memory” will give you a more detailed description of the working memory.

An embedded system without a control algorithm implemented memory is just like a new born baby. It is having all the peripherals but is not capable of making any decision depending on the situational as well as real world changes. The only difference is that the memory of a new born baby is self-adaptive, meaning that the baby will try to learn from the surroundings and from the mistakes committed. For embedded systems it is the responsibility of the designer to impart intelligence to the system.

In a controller-based embedded system, the controller may contain internal memory for storing the control algorithm and it may be an EEPROM or FLASH memory varying from a few kilobytes to megabytes. Such controllers are called controllers with on-chip ROM, e.g. Atmel AT89C51. Some controllers may not contain on-chip memory and they require an external (off-chip) memory for holding the control algorithm, e.g. Intel 8031AH.

2.1 CORE OF THE EMBEDDED SYSTEM

Embedded systems are domain and application specific and are built around a central core. The core of the embedded system falls into any one of the following categories:

- (1) General Purpose and Domain Specific Processors
 - 1.1 Microprocessors
 - 1.2 Microcontrollers
 - 1.3 Digital Signal Processors
- (2) Application Specific Integrated Circuits (ASICs)
- (3) Programmable Logic Devices (PLDs)
- (4) Commercial off-the-shelf Components (COTS)

If you examine any embedded system you will find that it is built around any of the core units mentioned above.

LO 1 Identify the building blocks of a typical Embedded System

2.1.1 General Purpose and Domain Specific Processors

Almost 80% of the embedded systems are processor/controller based. The processor may be a microprocessor or a microcontroller or a digital signal processor, depending on the domain and application. Most of the embedded systems in the industrial control and monitoring applications make use of the commonly available microprocessors or microcontrollers whereas domains which require signal processing such as speech coding, speech recognition, etc. make use of special kind of digital signal processors supplied by manufacturers like, Analog Devices, Texas Instruments, etc.

2.1.1.1 Microprocessors A Microprocessor is a silicon chip representing a central processing unit (CPU), which is capable of performing arithmetic as well as logical operations according to a pre-defined set of instructions, which is specific to the manufacturer. In general, the CPU contains the Arithmetic and Logic Unit (ALU), control unit and working registers. A microprocessor is a dependent unit and it requires the combination of other hardware like memory, timer unit, and interrupt controller, etc. for proper functioning. Intel claims the credit for developing the first microprocessor unit *Intel 4004*, a 4bit processor which was released in November 1971. It featured 1K data memory, a 12bit program counter and 4K program memory, sixteen 4bit general purpose registers and 46 instructions. It ran at a clock speed of 740 kHz. It was designed for olden day's calculators. In 1972, 14 more instructions were added to the 4004 instruction set and the program space is upgraded to 8K. Also interrupt capabilities were added to it and it is renamed as *Intel 4040*. It was quickly replaced in April 1972 by *Intel 8008* which was similar to *Intel 4040*, the only difference was that its program counter was 14 bits wide and the 8008 served as a terminal controller. In April 1974 Intel launched the first 8 bit processor, the *Intel 8080*, with 16bit address bus and program counter, and seven 8bit registers (A-E,H,L: BC, DE, and HL pairs formed the 16bit register for this processor). *Intel 8080* was the most commonly used processors for industrial control and other embedded applications in the 1975s. Since the processor required other hardware components as mentioned earlier for its proper functioning, the systems made out of it were bulky and were lacking compactness.

Immediately after the release of *Intel 8080*, Motorola also entered the market with their processor, *Motorola 6800* with a different architecture and instruction set compared to 8080.

In 1976 Intel came up with the upgraded version of 8080 – *Intel 8085*, with two newly added instructions, three interrupt pins and serial I/O. Clock generator and bus controller circuits were built-in and the power supply part was modified to a single +5 V supply.

In July 1976 Zilog entered the microprocessor market with its *Z80 processor* as competitor to *Intel*. Actually it was designed by an ex-Intel designer, *Frederico Faggin* and it was an improved version of Intel's 8080 processor, maintaining the original 8080 architecture and instruction set with an 8bit data bus and a 16bit address bus and was capable of executing all instructions of 8080. It included 80 more new instructions and it brought out the concept of register banking by doubling the register set. *Z80* also included two sets of index registers for flexible design.

Technical advances in the field of semiconductor industry brought a new dimension to the microprocessor market and twentieth century witnessed a fast growth in processor technology. 16, 32 and 64 bit processors came into the place of conventional 8bit processors. The initial 2 MHz clock is now an old story. Today processors with clock speeds over 3.0 GHz are available in the market. More and more competitors entered into the processor market offering high speed, high performance and low cost processors for customer design needs.

Intel, AMD, Freescale, GLOBALFOUNDRIES, TI, Cyrix, NVIDIA, Qualcomm, MediaTek, etc. are the key players in the processor market. Intel still leads the market with cutting edge technologies in the processor industry.

Different instruction set and system architecture are available for the design of a microprocessor. Harvard and Von-Neumann are the two common system architectures for processor design. Processors based on

Harvard architecture contains separate buses for program memory and data memory, whereas processors based on Von-Neumann architecture share a single system bus for program and data memory. We will discuss more about these architectures later, under a separate topic. Reduced Instruction Set Computing (RISC) and Complex Instruction Set Computing (CISC) are the two common Instruction Set Architectures (ISA) available for processor design. We will discuss the same under a separate topic in this section.

2.1.1.2 General Purpose Processor (GPP) vs. Application-Specific Instruction Set Processor (ASIP)

A General Purpose Processor or GPP is a processor designed for general computational tasks. The processor running inside your laptop or desktop (Pentium 4/AMD Athlon, etc.) is a typical example for general purpose processor. They are produced in large volumes and targeting the general market. Due to the high volume production, the per unit cost for a chip is low compared to ASIC or other specific ICs. A typical general purpose processor contains an Arithmetic and Logic Unit (ALU) and Control Unit (CU). On the other hand, Application Specific Instruction Set Processors (ASIPs) are processors with architecture and instruction set optimised to specific-domain/application requirements like network processing, automotive, telecom, media applications, digital signal processing, control applications, etc. ASIPs fill the architectural spectrum between general purpose processors and Application Specific Integrated Circuits (ASICs). The need for an ASIP arises when the traditional general purpose processor are unable to meet the increasing application needs. Most of the embedded systems are built around application specific instruction set processors. Some microcontrollers (like Automotive AVR, megaAV from Atmel), system on chips, digital signal processors, etc. are examples for application specific instruction set processors (ASIPs). ASIPs incorporate a processor and on-chip peripherals, demanded by the application requirement, program and data memory.

2.1.1.3 Microcontrollers A Microcontroller is a highly integrated chip that contains a CPU, scratch pad RAM, special and general purpose register arrays, on chip ROM/FLASH memory for program storage, timer and interrupt control units and dedicated I/O ports. Microcontrollers can be considered as a super set of microprocessors. Since a microcontroller contains all the necessary functional blocks for independent working, they found greater place in the embedded domain in place of microprocessors. Apart from this, they are cheap, cost effective and are readily available in the market.

Texas Instrument's *TMS 1000* is considered as the world's first microcontroller. We cannot say it as a fully functional microcontroller when we compare it with modern microcontrollers. TI followed Intel's *4004/4040*, 4 bit processor design and added some amount of RAM, program storage memory (ROM) and I/O support on a single chip, thereby eliminated the requirement of multiple hardware chips for self-functioning. Provision to add custom instructions to the CPU was another innovative feature of TMS 1000. TMS 1000 was released in 1974.

In 1977 Intel entered the microcontroller market with a family of controllers coming under one umbrella named *MCS-48TM* family. The processors came under this family were *8038HL*, *8039HL*, *8040AHL*, *8048H*, *8049H*, and *8050AH*. *Intel 8048* is recognised as Intel's first microcontroller and it was the most prominent member in the *MCS-48TM* family. It was used in the original IBM PC keyboard. The inspiration behind *8048* was Fairchild's *F8* microprocessor and Intel's goal of developing a low cost and small size processor. The design of *8048* adopted a true Harvard architecture where program and data memory shared the same address bus and is differentiated by the related control signals.

Eventually Intel came out with its most fruitful design in the 8bit microcontroller domain—the *8051 family* and its derivatives. It is the most popular and powerful 8bit microcontroller ever built. It was developed in the 1980s and was put under the family MCS-51. Almost 75% of the microcontrollers used in the embedded domain were *8051 family* based controllers during the 1980–90s. *8051* processor cores are used in more than

*MCS-48TM is a trade mark owned by Intel

100 devices by more than 20 independent manufacturers like Maxim, NXP, Atmel, etc. under the license from Intel. Due to the low cost, wide availability, memory efficient instruction set, mature development tools and Boolean processing (bit manipulation operation) capability, *8051 family* derivative microcontrollers are much used in high-volume consumer electronic devices, entertainment industry and other gadgets where cost-cutting is essential.

Another important family of microcontrollers used in industrial control and embedded applications is the **PIC** family micro controllers from Microchip Technologies (It will be discussed in detail in a later section of this book). It is a high performance RISC microcontroller complementing the CISC (Complex Instruction Set Computing) features of *8051*. The terms RISC and CISC will be explained in detail in a separate heading.

Some embedded system applications require only 8bit controllers whereas some embedded applications requiring superior performance and computational needs demand 16/32bit microcontrollers. Infineon, Freescale, Philips, Atmel, Maxim, Microchip etc. are the key suppliers of 16bit microcontrollers. Philips tried to extend the *8051* family microcontrollers to use for 16bit applications by developing the Philips XA (eXtended Architecture) microcontroller series.

8bit microcontrollers are commonly used in embedded systems where the processing power is not a big constraint. As mentioned earlier, more than 20 companies are producing different flavours of the *8051* family microcontroller. They try to add more and more functionalities like built in SPI, I2C serial buses, USB controller, ADC, Networking capability, etc. So the competitive market is driving towards a one-stop solution chip in microcontroller domain. High processing speed microcontroller families like ARM Cortex M Series are also available in the market, which provides solution to applications requiring hardware acceleration and high processing capability.

Freescale, Renesas, Zilog, Cypress (Spansion), Infineon, ST Micro Electronics, EPSON, Texas Instruments, Toshiba, NXP, Microchip, Analog Devices, Daewoo, Intel, Maxim, Sharp, Silicon Laboratories, HOLTEK, LAPIS, CYROD, Atmel, etc. are the key players in the microcontroller market. Of these Atmel has got special significance. They are the manufacturers of a variety of Flash memory based microcontrollers. They also provide In-System Programmability (which will be discussed in detail in a later section of this book) for the controller. The Flash memory technique helps in fast reprogramming of the chip and thereby reduces the product development time. Atmel also provides another special family of microcontroller called AVR (it will be discussed in detail in a later chapter), an 8bit RISC Flash microcontroller, fast enough to execute powerful instructions in a single clock cycle and provide the latitude you need to optimise power consumption.

The instruction set architecture of a microcontroller can be either RISC or CISC. Microcontrollers are designed for either general purpose application requirement (general purpose controller) or domain- specific application requirement (application specific instruction set processor). The *Intel 8051* microcontroller is a typical example for a general purpose microcontroller, whereas the automotive AVR microcontroller family from Atmel Corporation is a typical example for ASIP specifically designed for the automotive domain.

2.1.1.4 Microprocessor vs Microcontroller The following table summarises the differences between a microcontroller and microprocessor.

Microprocessor	Microcontroller
A silicon chip representing a central processing unit (CPU), which is capable of performing arithmetic as well as logical operations according to a predefined set of instructions	A microcontroller is a highly integrated chip that contains a CPU, scratchpad RAM, special and general purpose register arrays, on chip ROM/FLASH memory for program storage, timer and interrupt control units and dedicated I/O ports

(Contd.)

Microprocessor	Microcontroller
It is a dependent unit. It requires the combination of other chips like timers, program and data memory chips, interrupt controllers, etc. for functioning	It is a self-contained unit and it doesn't require external interrupt controller, timer, UART, etc. for its functioning
Most of the time general purpose in design and operation	Mostly application-oriented or domain-specific
Doesn't contain a built in I/O port. The I/O port functionality needs to be implemented with the help of external programmable peripheral interface chips like 8255	Most of the processors contain multiple built-in I/O ports which can be operated as a single 8 or 16 or 32 bit port or as individual port pins
Targeted for high end market where performance is important	Targeted for embedded market where performance is not so critical (At present this demarcation is invalid)
Limited power saving options compared to microcontrollers	Includes lot of power saving features

2.1.1.5 Digital Signal Processors Digital Signal Processors (DSPs) are powerful special purpose 8/16/32 bit microprocessors designed specifically to meet the computational demands and power constraints of today's embedded audio, video, and communications applications. Digital signal processors are 2 to 3 times faster than the general purpose microprocessors in signal processing applications. This is because of the architectural difference between the two. DSPs implement algorithms in hardware which speeds up the execution whereas general purpose processors implement the algorithm in firmware and the speed of execution depends primarily on the clock for the processors. In general, DSP can be viewed as a microchip designed for performing high speed computational operations for 'addition', 'subtraction', 'multiplication' and 'division'. A typical digital signal processor incorporates the following key units:

Program Memory Memory for storing the program required by DSP to process the data.

Data Memory Working memory for storing temporary variables and data/signal to be processed.

Computational Engine Performs the signal processing in accordance with the stored program memory. Computational Engine incorporates many specialised arithmetic units and each of them operates simultaneously to increase the execution speed. It also incorporates multiple hardware shifters for shifting operands and thereby saves execution time.

I/O Unit Acts as an interface between the outside world and DSP. It is responsible for capturing signals to be processed and delivering the processed signals.

Audio video signal processing, telecommunication, and multimedia applications are typical examples where DSP is employed. Digital signal processing employs a large amount of real-time calculations. Sum of products (SOP) calculation, convolution, fast fourier transform (FFT), discrete fourier transform (DFT), etc, are some of the operations performed by digital signal processors.

Blackfin®[†] processors from Analog Devices is an example of DSP which delivers breakthrough signal-processing performance and power efficiency while also offering a full 32-bit RISC MCU programming model. Blackfin processors present high-performance, homogeneous software targets, which allows flexible resource allocation between hard real-time signal processing tasks and non real-time control tasks. System control tasks can often run in the shadow of demanding signal processing and multimedia tasks.

2.1.1.6 RISC vs. CISC Processors/Controllers The term RISC stands for Reduced Instruction Set Computing. As the name implies, all RISC processors/controllers possess lesser number of instructions,

[†]Blackfin® is a Registered trademark of Analog Devices Inc.